Solution Requirements

Date	4 July 2026
Team ID	SWTID-2026-4860
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:-

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

● Functional Requirements (FR):- Functional requirements describe the main features the Edu Genie Google Powered Learning Assistant must provide to help users breakdown topics, answer academic queries , generate dynamic quizzes, and recommend structured roadmaps.

● Non-Functional Requirements (NFR):- Non-functional requirements describe how the EduGenie Google Gemini Powered Learning Assistant should perform by ensuring secure operation, processing speed, modular scalability, and a highly responsive user interface layout.

● Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1.	Authentication	Users can securely register an account and log in using verified email and password profiles before accessing the AI learning tools and dashboard workspaces.	High
2.	Authorization Levels	Registered student profiles can manage personal history states, view dynamic scores, and configure personal preference layouts. Admin accounts can monitor global system parameters.	High
3.	External Interfaces	The application securely integrates with the Google Gemini SDK for model text generation and connects to a localized SQLite backend instance for persisting structured table items.	High

4.	Transactions Processing	The system receives complex textbook strings or queries, maps the intent, builds structured context arrays, forwards payloads to the AI model, and returns response blocks.	High
5.	Reporting	The system renders real-time structured output blocks including automated multiple-choice assessment arrays, execution success summaries, and sequenced beginner-to-advanced roadmap trees.	Medium
6.	Business Rules	AI outputs and quiz variations are initiated solely after checking the validation states of the topic entry. Core educational feedback remains tailored strictly to current student performance thresholds.	High
7.	Compliance to Laws or Regulations	The application isolates private user profiles, enforces credential encryption rules, avoids hardcoding access variables, and conforms to basic consumer data protection protocols.	High
8.	Other	Students can navigate past historical topic generation tracks, reload previous roadmaps, and re-attempt older evaluation quizzes to evaluate progress.	Medium

Step 2: Non-Functional Requirements (NFR)

S.NO	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1.	Performance & Speed	The core framework router should capture payload arrays and fetch text answers back from the Gemini engine with minimal UI lag.	Response generation completed within 3 seconds.
2.	Scalability	The backend processing paths must handle multiple concurrent user threads executing queries simultaneously without infrastructure blockages.	Efficiently supports 100+ concurrent active requests.
3.	Security & Data Privacy	User structural parameters, salted password strings, and history logs must be isolated behind secure validation layers.	Encrypted user profiles and isolated local table models.

4.	Reliability & Availability	The local deployment runtime environment must execute continuous uptime cycles with built-in route fallback logic.	99% operational uptime availability.
5.	Usability & Accessibility	The single-page web view panel should provide an interactive responsive UI, avoiding heavy layout clusters.	Student can trigger full learning actions in 3 simple taps.
6.	Other	The modular codebase should feature separated front-end static components and discrete backend routing files.	Clean object-oriented structure for seamless future AI upgrades.